

User Behavior & Funnel Analysis

Google Merchandise Store — GA4 Demo Account

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1. Introduction

This case study analyses real user behavior data from the Google Merchandise Store using the Google Analytics 4 (GA4) Demo Account. The objective is to evaluate acquisition channel performance, map user engagement patterns across key pages, and identify drop-off points in the purchase funnel — culminating in data-backed recommendations to improve conversion rates.

The analysis covers three core areas: Traffic Acquisition, Page Engagement, and Funnel Performance.

2. Acquisition Analysis

2.1 Top Traffic Channels

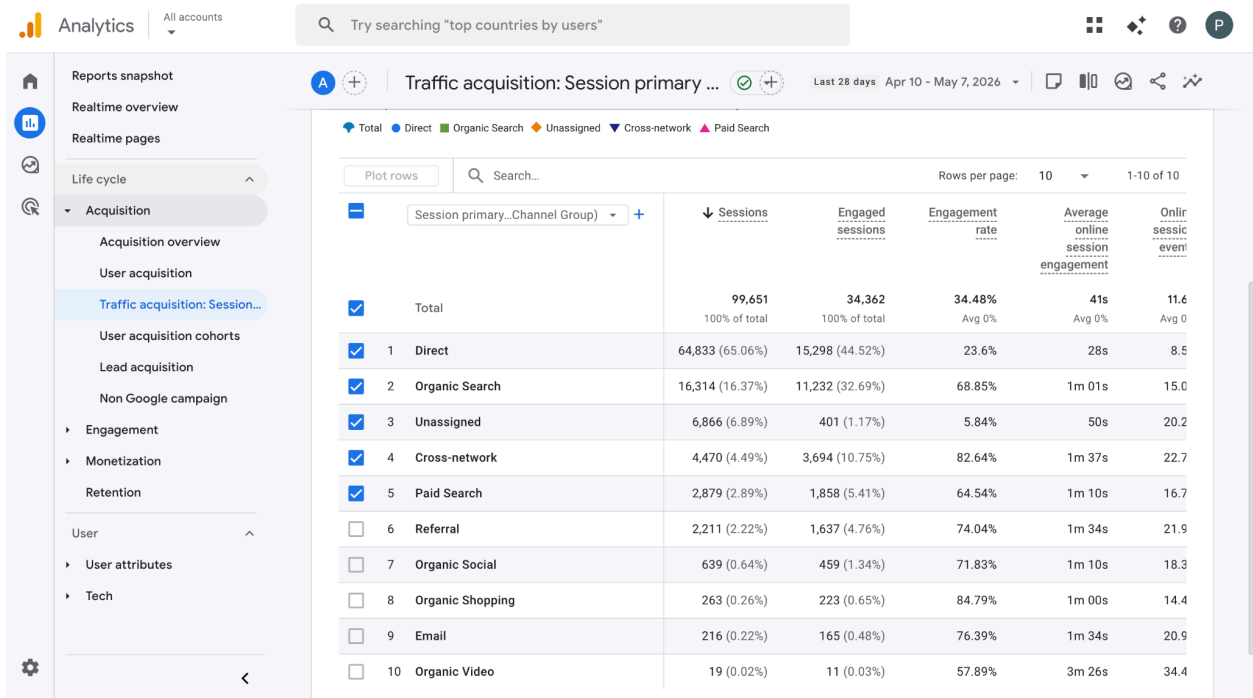
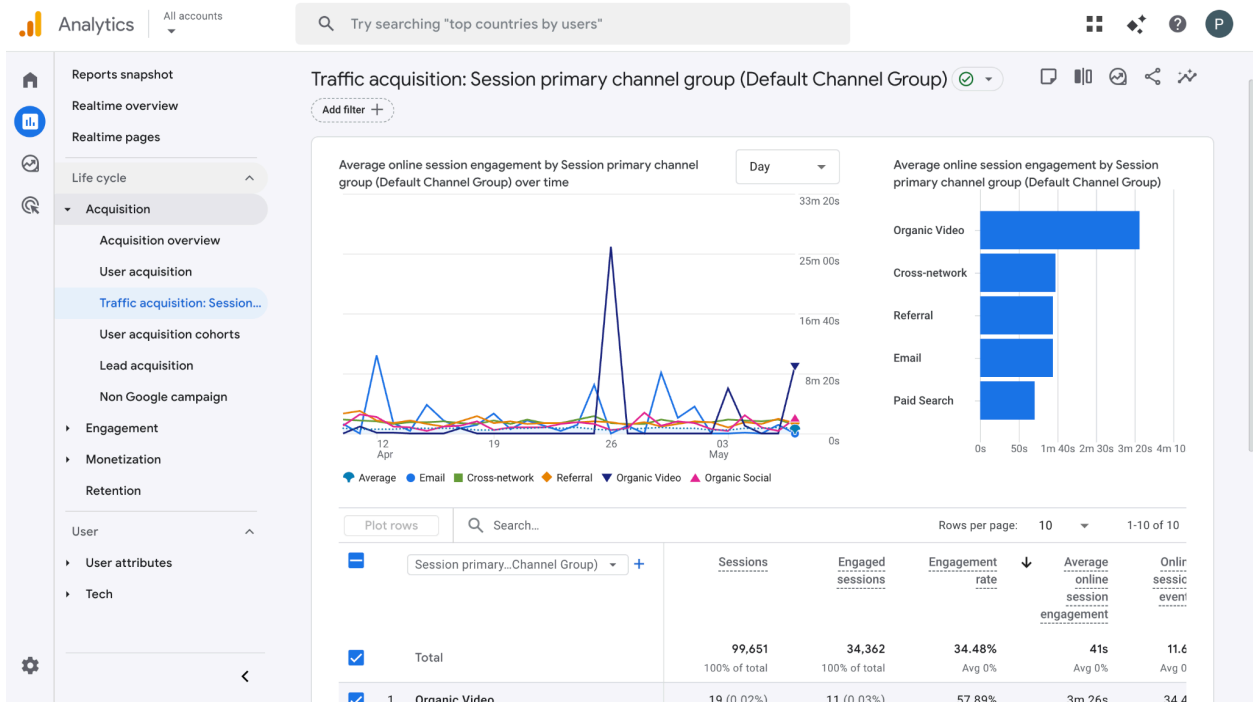
Using the Traffic Acquisition report in GA4, three primary channels were identified as the main drivers of sessions:

Channel	Traffic Volume	Conversion Quality
Direct	Highest	Moderate
Organic Search	High	Strong
Cross-network	Moderate	Strong

Key Insight	Direct traffic generated the highest session volume, indicating strong brand familiarity and a high proportion of returning visitors. However, Organic Search and Cross-network channels outperformed Direct on session key event rate — demonstrating that traffic volume and conversion quality are independent metrics.
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2.2 Strategic Implication

Optimising for high-volume channels alone is insufficient. A more effective strategy would focus on scaling the channels that convert well (Organic Search, Cross-network) while simultaneously improving the on-site experience for Direct visitors to close the conversion gap.



3. Engagement Analysis

3.1 Most Engaged Pages

The Pages and Screens report was used to map engagement across the website. A notable pattern emerged between page views and time-on-page:

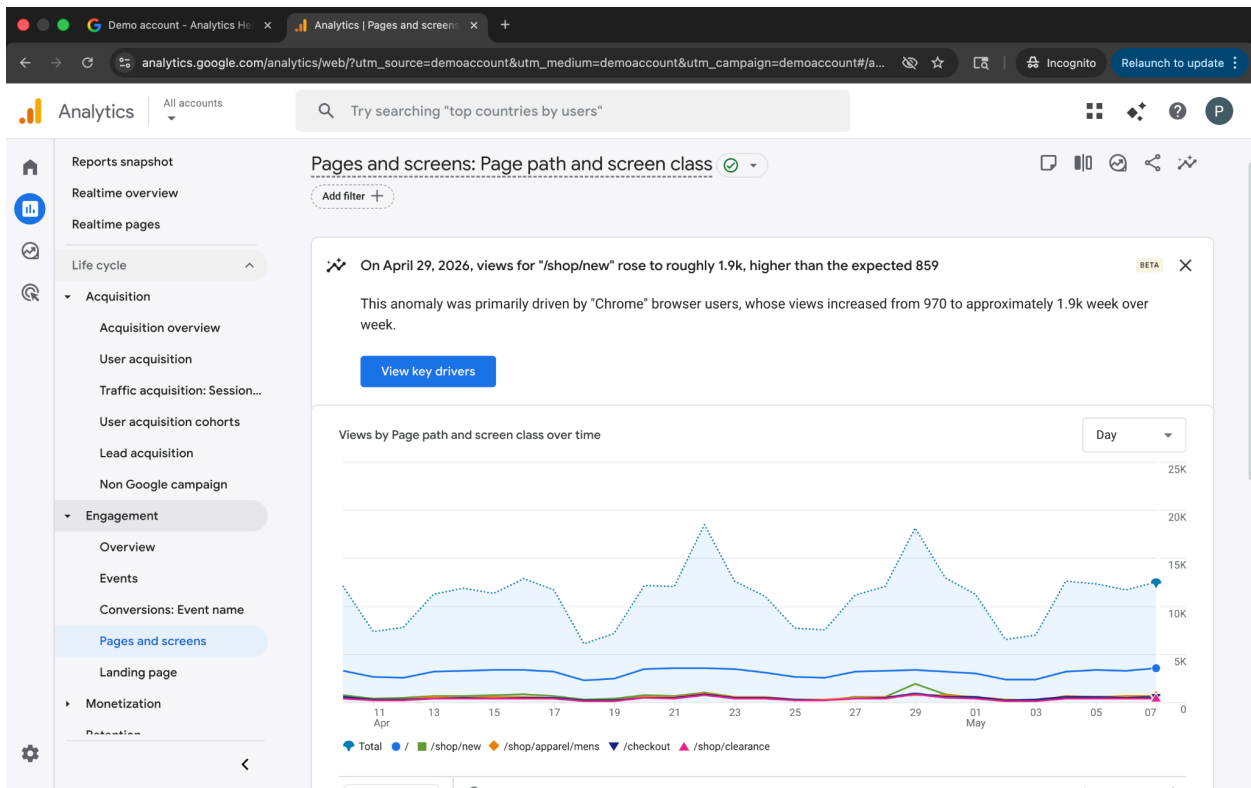
Page	View Volume	Avg. Engagement Time
Homepage (/)	Highest	Low
/shop/new	High	High
/shop/apparel/mens	High	High
/checkout	Lower	Highest

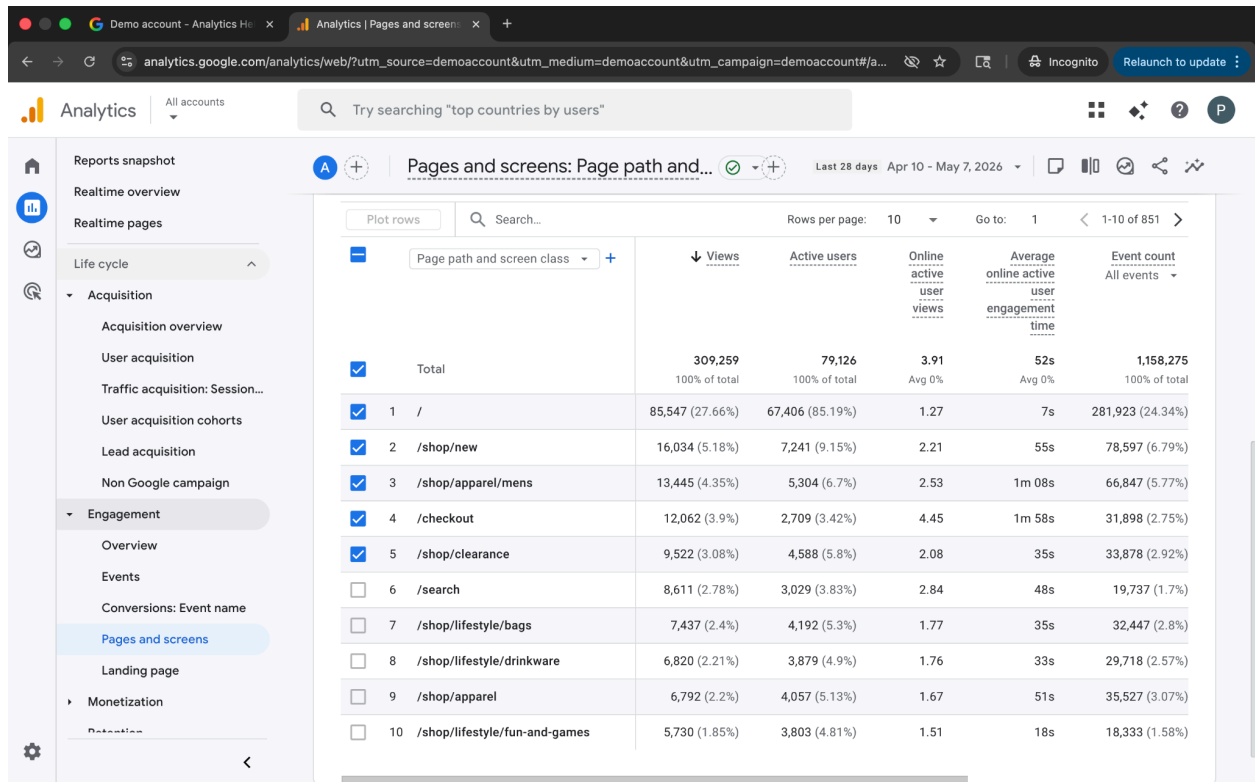
Key Insight

The homepage drives the most traffic but lowest engagement time — users are passing through, not converting. Product pages and checkout show the opposite: lower traffic but significantly higher engagement, indicating strong purchase intent among users who reach these pages.

3.2 Strategic Implication

The homepage is functioning primarily as a transit point rather than a conversion driver. Improving homepage personalisation — such as surfacing trending or recommended products — could redirect more users toward high-intent pages like /shop/new and reduce early drop-off.





4. Funnel Analysis

4.1 Funnel Design

A custom Funnel Exploration was built in GA4 using the following standard ecommerce progression:

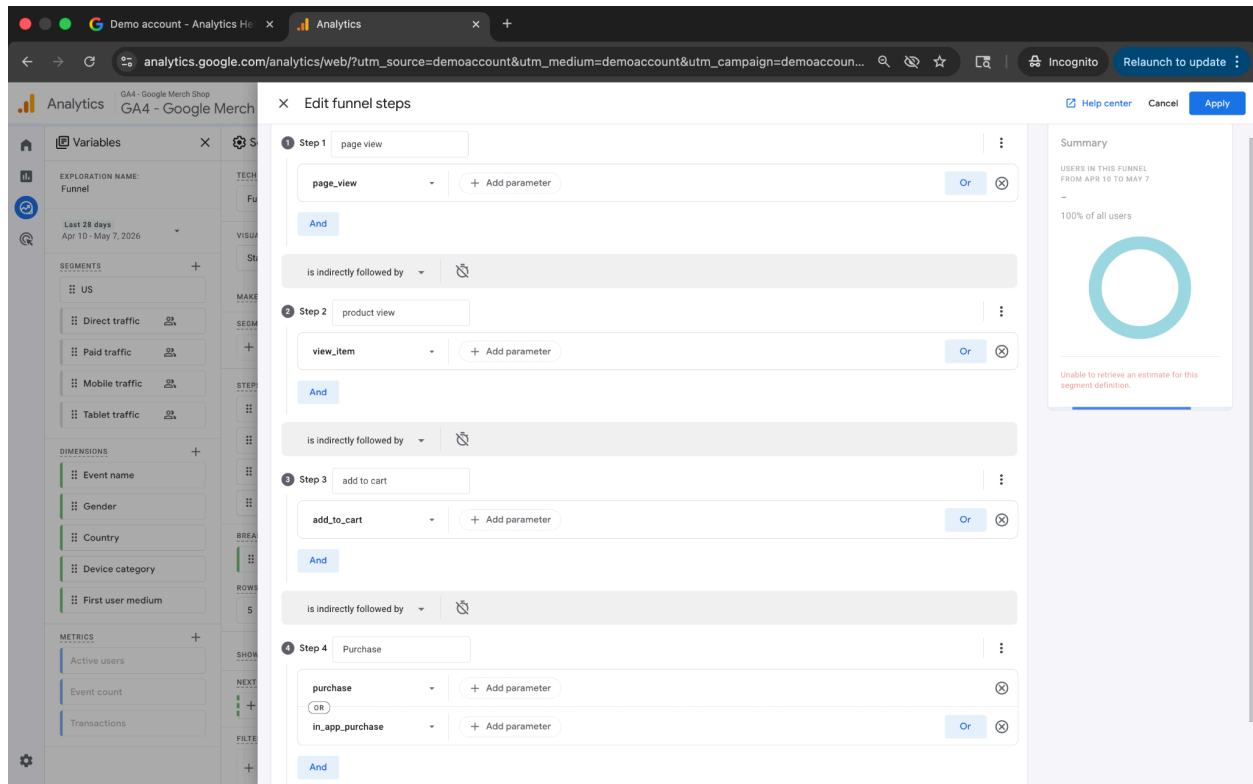
Stage	GA4 Event	What it Measures
1	view_item	User views a product page
2	add_to_cart	User adds item to cart
3	purchase	User completes checkout

4.2 Drop-off Analysis

Critical Finding	The highest drop-off point occurs between view_item and add_to_cart. A large proportion of users browse products but do not take the next step — indicating a significant gap between browsing intent and purchase intent.
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This pattern is consistent with a broader phenomenon in ecommerce known as 'browse abandonment'. Possible causes include:

- Insufficient product information — users can't find the details they need to feel confident purchasing
- Lack of social proof — no visible reviews, ratings, or user-generated content to build trust
- Price hesitation — no incentive to act immediately (no urgency, no offer)
- UX friction — the add-to-cart action may not be prominent enough on mobile or desktop



5. Recommendations

- 01 Enrich Product Pages to Reduce Browse Abandonment**

Add detailed product descriptions, high-quality images, size guides, and visible customer reviews directly on product pages. The data shows strong engagement time on product pages — users are spending time there but not converting. Better information can bridge this gap.
- 02 Simplify and Streamline the Checkout Flow**

Audit the steps between `add_to_cart` and `purchase`. Reduce the number of form fields, add progress indicators, and offer guest checkout. Any additional friction at this stage risks losing users who have already demonstrated purchase intent.
- 03 Launch Remarketing Campaigns Targeting Browse Abandoners**

Use GA4 audience segments to retarget users who triggered `view_item` but not `add_to_cart`. Cross-network channels already show strong conversion quality — remarketing via these channels with product-specific ads can re-engage high-intent users at a lower acquisition cost.

Improve Homepage Personalisation to Drive Deeper Funnel Entry

Since the homepage has the highest views but lowest engagement time, introduce personalised product recommendations or 'trending now' sections. Reducing the number of clicks between homepage and high-intent pages like /shop/new can meaningfully improve funnel entry rates.

6. Conclusion

This GA4 analysis of the Google Merchandise Store reveals three clear opportunity areas: acquisition channel optimisation, product page conversion improvement, and funnel drop-off reduction.

The core insight is that traffic volume is not a reliable proxy for conversion performance. Channels like Organic Search and Cross-network outperform Direct on conversion quality despite lower volume — suggesting the business should prioritise channel quality over channel scale.

The most actionable finding is the high drop-off between product view and add-to-cart. Addressing this single stage — through richer product content, social proof, and a frictionless checkout — represents the highest-impact opportunity to improve overall revenue performance.

Analysis conducted using GA4 Demo Account | Google Merchandise Store | P. Sai Smrithi